

SEC P vs NP log 2026-05-13

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-13 01:00 UTC

SEC P vs NP — daily log 2026-05-13

3 cycles recorded

Real Radical Dimension Inverse Proportional to SOS Degree for Max-CUT

- **Verdict:** INCONCLUSIVE
- **Bridge:** Real Algebraic Geometry $\times$ SOS Degree for Max-CUT
- **Recorded:** 2026-05-13 00:19 UTC
- **Entry ID:** d23b11b813ab

Statement

For any Max-CUT instance with n variables, the dimension of the real radical of the ideal generated by its constraints is $\Theta(\log n)$, implying the SOS degree required to refute it is $\Omega(n^{\{1/2\}})$

Rationale

The real radical captures the algebraic structure of the feasible region for Max-CUT. Its dimension reflects the complexity of the variety, which directly influences the SOS hierarchy's required degree to capture its geometry.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.05s

```
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```

TRIAL: {'metric_name': 'dimension_ratio', 'metric_value': 1.0, 'instances_tested': 1, 'conjecture_holds': True}
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RESULT: SUPPORTED mean=1.0 std=0.0 support_fraction=1.0

```

Judge reasoning

All trials tested $n=1$ instances, making the metric trivial and conjecture vacuously true. Selection bias likely misses adversarial cases where dimension diverges from 1.0. | next: Test on larger n (e.g., $n=1000$) with randomized instance generators to validate metric scaling

Real Variety Connected Components Lower Bound SOS Degree for Max-CUT

- **Verdict:** INCONCLUSIVE
- **Bridge:** Real Algebraic Geometry $\times$ SOS Degree for Max-CUT
- **Recorded:** 2026-05-13 00:30 UTC
- **Entry ID:** 1ea2045ec8f6

Statement

For a random Max-CUT instance on n variables, the SOS degree required to refute it is at least the number of connected components of the real variety defined by the instance's constraints.

Rationale

The structure of the real variety's connected components reflects the complexity of the feasible region, which may necessitate higher SOS degrees to capture the necessary constraints.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.03s

```

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TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 12, 'instances_tested': 1, 'conjecture_holds': True}
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RESULT: SUPPORTED mean=7.633333333333334 std=2.7016455890602837 support_fraction=1.0

```

Judge reasoning

Safety rail: critic_challenge | original judge: All 15 trials confirmed the conjecture holds with no counterexamples found. The support fraction (1.0) exceeds the pre-registered threshold of 80%. | next: Test larger instances with varying connectivity structures to validate across more diverse cases

Plethysm Coefficient Gap in Symmetric Powers of Permanent vs Determinant

- **Verdict:** INCONCLUSIVE
- **Bridge:** Representation Theory of Symmetric Groups (Plethysm Coefficients) $\times$ Boolean Circuit Size for Permanent Polynomial
- **Recorded:** 2026-05-13 00:43 UTC
- **Entry ID:** f3dc4c5f9c43

Statement

For any CNF formula Φ with n variables, the plethysm coefficient of the permanent polynomial ($\lambda \vdash n$) exceeds that of the determinant polynomial by a factor of $\Omega(2^{\{n/2\}})$, where λ is the Young diagram corresponding to Φ 's clause structure.

Rationale

Plethysm coefficients encode decomposition complexity of symmetric tensors; the permanent's higher coefficients suggest intrinsic structural complexity that resists efficient circuit simulation, aligning with GCT's orbit closure separation goals.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_bd90514d.py", line 88, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_bd90514d.py", line 61, in run_trial
    lambda_diag = [len(list(g)) for _, g in itertools.groupby(sorted(formula), key=lambda x: len(x))]
                  ^^^^^^^^^
```

NameError: name 'itertools' is not defined. Did you forget to import 'itertools'?

Judge reasoning

Test crashed due to missing itertools import, preventing data collection | next: Fix itertools import in test code and re-run experiments